

Problem Statement: Audio Signal Processing Assistant Agent

Audio signal processing circuits, including amplifiers, filters, and preamplifiers, play a crucial role in communication systems, audio equipment, and modern consumer electronics. During the design and troubleshooting process, engineers and students often face challenges such as unwanted noise, signal distortion, grounding faults, and improper filter design.

Identifying and resolving these issues requires strong expertise in signal processing principles and circuit analysis. However, obtaining relevant information often involves consulting multiple technical resources, manuals, and online references, making the process time-consuming and complex.

The objective is to develop an intelligent AI-based assistant that can analyze audio circuit problems, provide accurate diagnostics, explain underlying signal processing concepts, and recommend effective design improvements and troubleshooting solutions.

The Objective

Build an **AI-powered Audio Signal Processing Assistant** that provides intelligent support for engineers, students, and electronics enthusiasts in analyzing, troubleshooting, and improving audio signal processing circuits. The solution should:

Audio Circuit Diagnosis – Detect and analyze issues such as noise, distortion, grounding faults, signal interference, and incorrect amplifier or filter configurations.

Signal Processing Knowledge RAG Retrieval – Retrieve and summarize relevant information from technical manuals, research papers, datasheets, and signal processing resources to provide accurate explanations and solutions.

Circuit Design Guidance – Recommend improvements in amplifier design, filter selection, component values, gain settings, and noise reduction techniques.

Audio Performance Analysis – Interpret user-provided circuit parameters, schematics, signal measurements, or error descriptions and provide real-time feedback.

Multi-Agent System –

- **Audio Knowledge Agent** (retrieves signal processing concepts and technical references)
- **Circuit Diagnosis Agent** (identifies faults and root causes of audio issues)
- **Design Optimization Agent** (suggests circuit improvements and component recommendations)
- **Signal Analysis Agent** (analyzes frequency response, noise levels, and distortion characteristics)

Visualization Dashboard – Display signal waveforms, frequency spectra, filter responses, noise analysis, and recommended design modifications for better understanding and decision-making.

Proposed Solution -

The proposed solution is a multi-agent AI-powered Audio Signal Processing Assistant built using LangFlow, IBM watsonx.ai, and Granite Models to provide intelligent, real-time support for analyzing, troubleshooting, and optimizing audio circuits. The system addresses common challenges in audio signal processing, including noise, distortion, grounding issues, improper filter configurations, and amplifier performance limitations through a modular AI-driven architecture.

At its core, the solution utilizes a Retrieval-Augmented Generation (RAG) pipeline to retrieve accurate technical knowledge from trusted sources such as audio engineering manuals, electronic component datasheets, signal processing textbooks, and research publications. This information is converted into embeddings and stored in a vector database, allowing Granite models to retrieve, understand, and summarize relevant technical information for user queries.

The system consists of four key agents:

Audio Knowledge Agent retrieves and summarizes signal processing concepts, circuit theories, component specifications, and technical references using the RAG pipeline.

Circuit Diagnosis Agent analyzes user-reported issues such as noise, distortion, signal loss, grounding faults, and unstable amplifier behavior, and provides possible root causes and troubleshooting steps.

Design Optimization Agent recommends improvements in amplifier circuits, filter design, gain settings, component selection, and noise reduction techniques to enhance overall audio performance.

Signal Analysis Agent interprets input signal parameters, frequency responses, waveform characteristics, and distortion measurements to provide detailed analysis and performance recommendations. The solution includes a user-friendly dashboard built using Streamlit or React that visualizes audio waveforms, frequency spectra, filter responses, noise characteristics, and suggested design improvements for easier understanding and decision-making. LangFlow is used to orchestrate multi-agent workflows, while IBM watsonx.ai provides scalable AI model deployment, embeddings, and responsible AI capabilities. Granite models enable advanced reasoning, technical summarization, and contextual assistance. Overall, the proposed solution delivers an end-to-end intelligent audio engineering assistant that simplifies circuit debugging, enhances learning, accelerates design optimization, and improves the overall efficiency of audio signal processing development.